

Manual

PCC Module Euromap E63 Communication PCC-E63 8.1

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1 Overview PCC Module Euromap E63 Communication

Purpose

The E63 Communication protocol is often used to communicate between machines and application programs.

The Euromap E63 Machine Interface

E63 is a file interface and the format is defined in the Euromap 63 specification.

See also <http://www.euromap.org>

The communication between server (HYDRA terminal or collection server) and machine takes place via an interconnected device. Files are transferred via a network connection using FTP or another protocol to access files in a server. HYDRA accesses these files per Windows file access locally or via the network depending on the configuration.

Latest version of PCC module for HYDRA is the E63 V1.05a (<http://www.euromap.org/downloads/eu63.pdf>).

This access path can be used for data from HYDRA modules:

- MDE - Machine Data Collection (counters/meters, cycle, status)
- PDV - Process Data Collection (process values)
- DNC - Direct Numerical Control (setting values)
- Scale values

Other data (i.e. MPL or ADE data) and more complex form of communication can be processed by extending the PCC-ADP application adapter.

Implementation notes

You use the function package if:

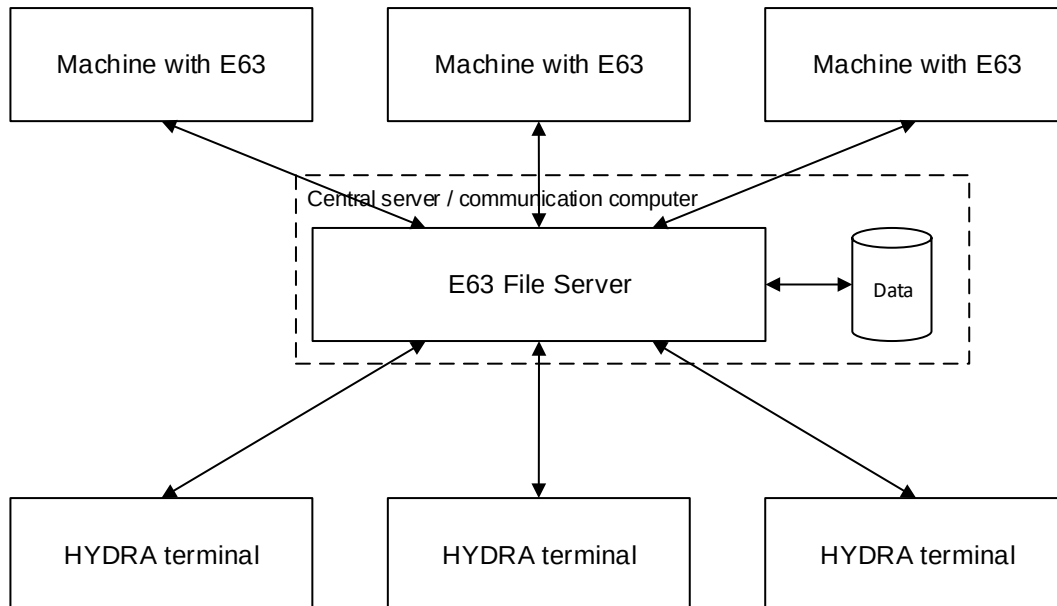
- You want to connect a machine to the MES using the E63 interface.
- The modules required to implement the protocol must exist or be established specifically to the machine. Specifications are defined during machine installation service.

Integration

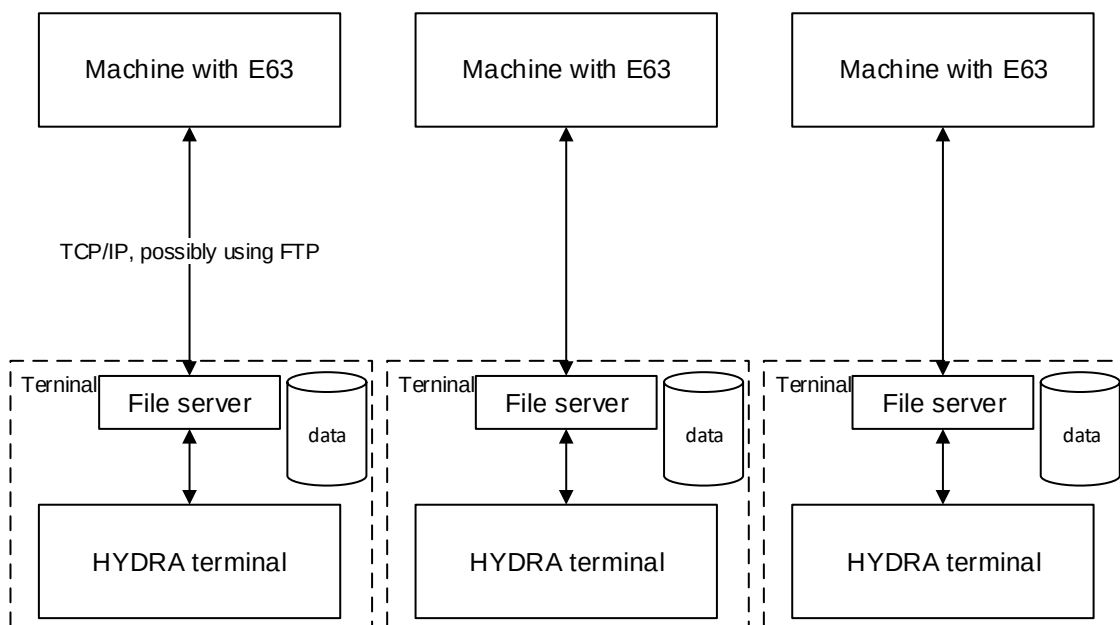
The protocol module is integrated in the HYDRA-PCC. Please also see the documents dealing with the products [SCS-PCP](#) and [SCS-PCB](#).

System configuration using HYDRA Euromap

E63 Configuration 1: Communication using central server



E63 Configuration 2: Direct Communication with Hydra terminals



14.1 Requirements

File transfer between machine control and the superordinate computer (i.e. communication device) takes place via the Ethernet interface. There is a screen in the control to set the communication parameters (like the TCP/IP address of the superordinate computer and the name of the "session" directory). HYDRA processes the files using Windows file access via the operating system locally or via the network.

Required are:

--> Network hardware

--> Communication device - might be identical to the HYDRA terminal

--> Network communication software (file server software and network protocol) for machine access and also HYDRA file access

Functions

- An interface module is required per connected machine, aggregate or control. This module assists the transfer of process and machine data from the control of machines and also for the transfer of setting parameters (NC data) to the control using Euromap E63 interfaces.

2 PCC-E63

2.1 General

EUROMAP 63 or E63 is a standardized file-based interface according to the Euromap63 specification, which is used to exchange data with injection molding machines via FTP server or network drive.

The communication between server (HYDRA terminal or shop floor server) and machine is performed via file transfer or communication directory. The network (TCP/IP) and the FTP protocol or another protocol are used to access the file on the communication server for the file transfer from the machine. Depending on the configuration, HYDRA accesses these files via Windows file access locally or via the network.



The HYDRA protocol module E63 is based on the E63 version V1.05. See also <http://www.euromap.org/downloads/eu63.pdf>



The EUROMAP63 interface is not real time capable. The data transfer time is sometimes longer than the time required for specific partial cycles of the production process. This is due to the architecture of the file interface of the EUROMAP63 standard.

2.2 Purpose

To connect machines to the HYDRA system, HYDRA provides an EUROMAP63 client (PCC-E63) integrated in the software. This client has been realized as protocol module in the HYDRA-PCC (Process Communication Controller). You can use the client to read all data fields (tabs) of the machine that are provided by the EUROMAP63 interface of the machine. This access can be used for data of the following HYDRA product groups:

- MDE (counter, cycle)
- PDV (process values)
- DNC (setting values)

2.3 Requirements

- The required software (PCC) must be installed on the PCC.
- The PCC has read and write permission in the directory with the E63 machine data (SESSION directory).
- The EUROMAP63 function must be activated for the machine.
- The connection test to the machine must be successful.

2.3.1 Required software

You require the software below to use the E63 driver:

File name	Description	Storage location
e63.dll	E63-MDE- und PDV-Treiber	Default: AIP directory
e63.bsp	Example configuration MDE, PDV	Default: AIP directory
e63.ini	Machine-specific E63 configuration	Default: AIP directory
e63_dnc.dll	E63-DNC-driver	Default: AIP directory
e63_dnc.bsp	Example configuration DNC	Default: AIP directory
e63_dnc.ini	Machine-specific E63-DNC configuration	Default: AIP directory
getid.job	E63 job file for manual connection test	
sess0001.req	E63 request file for manual connection test	

2.3.2 Supported control types/injection molding machines

This list includes the supported control types of the injection molding machines for the PCC driver E63. If a control type is not included in this list, contact MPDV for advice.



It can sometimes be necessary to adjust the E63 driver for the machine after updates of the machine manufacturer. MPDV has no influence on the updates of the machine manufacturers.

Machine manufacturer	Control type	Connection via	Notes
Battenfeld	UNILOG B4 UNILOG B5	UNC access (PCC)	
DEMAG	NC4 NC5 NC6	UNC access (PCC) / FTP access (machine)	
Engel	CC100 CC200 CC300	UNC access (PCC) / Engel Middleware	Middleware of Engel is required
Fanuc		UNC access (PCC) / FTP access (machine)	
Ferromatik		UNC access (PCC) / FTP access (machine)	
Krauss-Maffei (KM)	MC4 MC5 MC6	UNC access (PCC) / FTP access (machine)	
STORK IMM		UNC access (PCC) / FTP access (machine)	

2.3.3 Connection test to E63 machine

With the EUROMAP63 standard, you can test the communication independently of the shop floor client (PCC). Below, MPDV provides a recommendation of how to test the machine communication.

The customer is responsible for the error-free communication between machine, middleware (FTP/file server, eFactory, etc.) and deployment directory of the E63 driver.



The recommended procedure for testing the E63 machine communication is general. The specific details depend on the technical realization of the machine manufacturer. Only the machine manufacturer can provide detailed information.

How to proceed

1. Check the error-free network communication between machine (usually FTP client), server (usually FTP server) and PCC. You can use common network tools like Ping, Tracert and ftp for the test.
2. When the network connection is guaranteed, you must test the data exchange between machine, server and PCC. Use the logs of the machine and the FTP server to find out whether the connection was successfully established.
3. Note the examples of the manufacturers.
4. Ensure that the Windows user, which is used to run the PCC, has read and write access for the directory where the EUROMAP63 files of the relevant machine are stored.
5. When the connection to the machine is established, you can manually test the data exchange between terminal and machine. Copy "getid.job" and "sess0001.req" to the transfer directory. The machine monitors the session directory and writes the parameters in a second file "getid.dat".

2.4 Integration in HYDRA

The protocol module is integrated in the HYDRA-PCC. For further information, refer to the manuals of the products [SCS-PCP](#) and [SCS-PCB](#).

2.5 Protocol modules PCC-E63

2.5.1 Technical details

The driver is available as a component in the PCC. Find below the technical details.

Component description	
Name:	E63
Class:	
Files:	e63.dll e63.ini
Available PCC channels	C:C001 P:P001 T:T001 I:I001 Z:Z001
Notes:	The driver e63.dll is used for the HYDRA product groups MDE and PDV.
Processing:	

Component description	
Name:	E63_DNC
Class:	
Files:	e63_dnc.dll e63_dnc.ini
Available PCC channels	D:DNC@MNR
Notes:	The driver e63.dll is used for the HYDRA product groups DNC.
Processing:	



There is no “automatic” mode for the PDV data collection. The measured value data must therefore be collected or recorded in relation to time or pieces (i.e. cyclically).

In addition, offline collection does not transfer time stamps for measured values.

The paragraph below describes the configurable ini. items and sections of the e63.ini and e63_dnc.ini.

There is always a section with the name "SYSTEM". Connections to the E63 machine (sessions) are defined using further sections. The other sections can have any name. Umlauts and special characters are NOT permitted.



Important: The parameters are case sensitive.
You must observe upper/lower case!

2.5.2 Section [service]

2.5.2.1 Info

Parameter description	
Name:	info
Data type:	String
Value range:	
Default:	e63.dll
Type:	Mandatory parameter
Description:	Driver name belonging to the ini file
Notes:	
Example	info=e63.dll
File	e63.ini, e63_dnc.ini

2.5.2.2 Interval

Parameter description	
Name:	intervall
Data type:	Integer
Value range:	
Default:	0
Type:	Mandatory parameter
Description:	
Notes:	
Example	intervall=0
File	e63.ini, e63_dnc.ini

2.5.2.3 Tracing

Parameter description	
Name:	Tracing
Data type:	Integer
Value range:	0..1
Default:	0
Type:	Mandatory parameter
Description:	Activates the tracing
Notes:	
Example	tracing=0
File	e63.ini, e63_dnc.ini

2.5.2.4 TraceLevel

Parameter description	
Name:	TraceLevel
Data type:	Integer
Value range:	
Default:	0
Type:	Mandatory parameter
Description:	Controls the level of detail of the tracing
Notes:	
Example	TraceLevel=0
File	e63.ini, e63_dnc.ini

2.5.3 Section [instance]

You can create communication instances within the driver. You usually define one instance per E63 machine. You can use any name for the sections (e.g. machine name).

2.5.3.1 loglevel

Parameter description	
Name:	loglevel
Data type:	Integer
Value range:	0-20
Default:	1

Type:	Mandatory parameter
Description:	Controls the level of detail of the log
Notes:	1=standard value; 20=service information that must absolutely be deactivated again to avoid run time problems.
Example	loglevel=1
File	e63.ini, e63_dnc.ini

2.5.3.2 qpath

Parameter description	
Name:	qpath
Data type:	String
Value range:	
Default:	
Type:	Mandatory parameter
Description:	Source path for JOB files of the E63 communication
Notes:	UNC paths are supported
Example	C:\MPDV\AIP2\cfg\ERGOTECH150x14\
File	e63.ini, e63_dnc.ini

2.5.3.3 sesspath

Parameter description	
Name:	sesspath
Data type:	String
Value range:	
Default:	
Type:	Mandatory parameter
Description:	Session path to the transfer directory of the machine for E63 communication
Notes:	UNC paths are supported
Example	\\FTPServer\E63\ERGOTECH150x14\
File	e63.ini, e63_dnc.ini

2.5.3.4 SAVE_COUNTER

Parameter description	
Name:	SAVE_COUNTER
Data type:	String
Value range:	ON/OFF
Default:	
Type:	Optional parameter
Description:	Counter values are saved in a separate file.
Notes:	
Example	SAVE_COUNTER=ON
File	e63.ini

2.5.3.5 GET_CURRENT_ALARMS

Parameter description	
Name:	GET_CURRENT_ALARMS
Data type:	String
Value range:	ON/OFF
Default:	OFF
Type:	Optional parameter
Description:	Receive alert messages via the job CetCurrentAlarms
Notes:	Only change after consultation with MPDV.
Example	GET_CURRENT_ALARMS=OFF
File	e63.ini

2.5.3.6 MACHINE_CYCLE

Parameter description	
Name:	MACHINE_CYCLE
Data type:	Integer
Value range:	01-59
Default:	
Type:	Mandatory parameter
Description:	Response interval of machine for MDE requests
Notes:	max. 59 sec. possible; value entry must be two digit

Example	MACHINE_CYCLE=15
File	e63.ini

2.5.3.7 OVERFLOW

Parameter description	
Name:	OVERFLOW
Data type:	Integer
Value range:	0-4294967295
Default:	
Type:	Mandatory parameter
Description:	Counter overflow of machine
Notes:	Machine-specific; is required for the MDE processing of counters. This parameter is used to define an overflow (tab) of the machine counter. A wrong setting can result in a wrong processing.
Example	OVERFLOW=999999
File	e63.ini

2.5.3.8 OVERFLOWMODE

Parameter description	
Name:	OVERFLOWMODE
Data type:	String
Value range:	0; 1; RESET
Default:	
Type:	Mandatory parameter
Description:	0 = in case of an overflow, the machine counter starts at "0" 1 = in case of an overflow, the machine counter starts at "1" RESET = the machine counter can be reset to zero during operation. Counting starts at "0"
Notes:	With injection molding machines, the counter is usually reset in the machine panel. In this case, the overflow is not integrated. Select the value RESET.
Example	OVERFLOWMODE = RESET
File	e63.ini

2.5.3.9 Machine type

Parameter description	
Name:	Machine type
Data type:	String
Value range:	DEMAG; KM; ENGEL; BATTENFELD
Default:	
Type:	Optional parameter
Description:	Selection of machine type
Notes:	
Example	Maschine type=DEMAG
File	e63.ini, e63_dnc.ini

2.5.3.10 E63 controller

Parameter description	
Name:	E63 controller
Data type:	String
Value range:	CCL912; CIPC; DIP12
Default:	
Type:	Optional parameter
Description:	Selection of E63 controller
Notes:	CCL912 (DEMAG) CIPC (DEMAG) DIP12 (DEMAG)
Example	E63-Controller=DIP12
File	e63.ini, e63_dnc.ini

2.5.3.11 Timeout

Parameter description	
Name:	Timeout
Data type:	Word [32bit]
Value range:	0-4294967295
Default:	20000

Type:	Mandatory parameter
Description:	Timeout for machine in ms
Notes:	The machine must respond within the time specified.
Example	Timeout=20000
File	e63.ini, e63_dnc.ini

2.5.3.12 Polltime

Parameter description	
Name:	Polltime
Data type:	Integer
Value range:	
Default:	1000
Type:	Optional parameter
Description:	Timer for the cyclic E63 processing in ms
Notes:	
Example	Polltime=1000
File	e63.ini

2.5.3.13 jobup

Parameter description	
Name:	jobup
Data type:	String
Value range:	
Default:	
Type:	Optional parameter
Description:	DNC upload
Notes:	
Example	jobup=UPLOAD
File	e63_dnc.ini

2.5.3.14 jobdown

Parameter description

Name:	jobdown
Data type:	String
Value range:	
Default:	
Type:	Optional parameter
Description:	DNC Download
Notes:	
Example	jobdown=DOWNLOAD
File	e63_dnc.ini

2.5.3.15 Zip

Parameter description	
Name:	Zip
Data type:	String
Value range:	ON/OFF
Default:	
Type:	Optional parameter
Description:	Zip several machine data to a DNC file package
Notes:	
Example	Zip=ON
File	e63_dnc.ini

2.5.3.16 DirzZip

Parameter description	
Name:	DirZipzipdir
Data type:	String
Value range:	ON/OFF
Default:	
Type:	Optional parameter
Description:	Zip machine data including folder to a DNC file package
Notes:	
Example	DirZip=ON
File	e63_dnc.ini

Machine-specific configuration:

Machine type	Controller type	Zip	DirZip
ENGEL			
CC100	CC100	ON	OFF
CC200	--	OFF	ON
CC300	CC300	OFF	ON
KraussMaffei			
MC5	--	ON	OFF
MC6	--	ON	OFF

2.5.4 Driver-specific channel options

Some channel options are available for this PCC driver. If required, they can be added to each channel configuration. The following sections describe these options.

The channels must always be unique in an INI file. For example, it is not permitted to use the channel C001 twice. Optionally, you can assign a machine to a channel using the @ character (e.g. C001@MNR12345). The combination channel + machine is then unique.

2.5.4.1 Counter channels [C]

Parameter description	
Name:	C:C001
Data type:	INTEGER
Value range:	C001-C999
Default:	
Type:	Mandatory parameter
Description:	Counter for the collection of quantities
Notes:	only E63 driver C:C010@EN110-1=ActCntCyc
Example	C:C001= ActCntCyc

2.5.4.2 Cycle channels [Z]

Parameter description	
Name:	Z:Z001
Data type:	String, no array
Value range:	Z001-Z999
Default:	
Type:	Optional parameter
Description:	Cycle channels to collect cycle times from machine
Notes:	E63 variable is automatically requested and transferred from the machine control. only E63 driver Z:Z010@EN110-1=ActTimCyc
Example	Z:Z001=ActTimCyc

2.5.4.3 Process data channels [P]

Parameter description	
Name:	P:P001
Data type:	String, no array
Value range:	P001-P999
Default:	
Type:	Optional parameter
Description:	Process data channel to collect process values
Notes:	
Example	P:P0001=ActPrsXfrHyd[1]

2.5.4.4 Trigger channels [T]

Parameter description	
Name:	T:T001
Data type:	String, no array
Value range:	Any
Default:	
Type:	Optional parameter
Description:	Trigger channels to trigger process data collection
Notes:	E63 variable is automatically requested and transferred from the machine control. Only E63 driver Is required for a triggered process data collection when triggered via the E63 connection.
Example	T:T001=ActCntCyc

2.5.4.5 DNC channels [D]

Parameter description	
Name:	D:DNC_MNR000=D:DNC_ + machine name
Data type:	String, no array
Value range:	Any
Default:	
Type:	Optional parameter
Description:	DNC channel to transfer setting data records
Notes:	
Example	D:DNC_500001=D:DNC_ + machine1

3 Configuration

3.1 Overview of file assignment

In the local HYDRA-AIP or HYDRA-PCC directory, you must enter the drivers "e63.dll" and "e63_dnc.dll" in the pccdll.ini (only required with DNC).

```
[SERVICE]
tracing=0
;DNC_BLOCKED_PDV=ON
;DNC_TIMEOUT=240
;BYPASS_PDV=ON
DLL-VERSION=Version 7.2.2.23 23.02.2017

[DRIVER_1]
driver=e63.dll

[DRIVER_2]
driver=e63_dnc.dll
```

Image pccdll.ini

When you have activated the driver in the pccdll.ini, you must configure the drivers e63.dll and e63_dnc.dll. To facilitate configuration, MPDV provides two sample configurations (e63.bsp and e63_dnc.bsp).

3.2 E63 MDE and PDV configuration

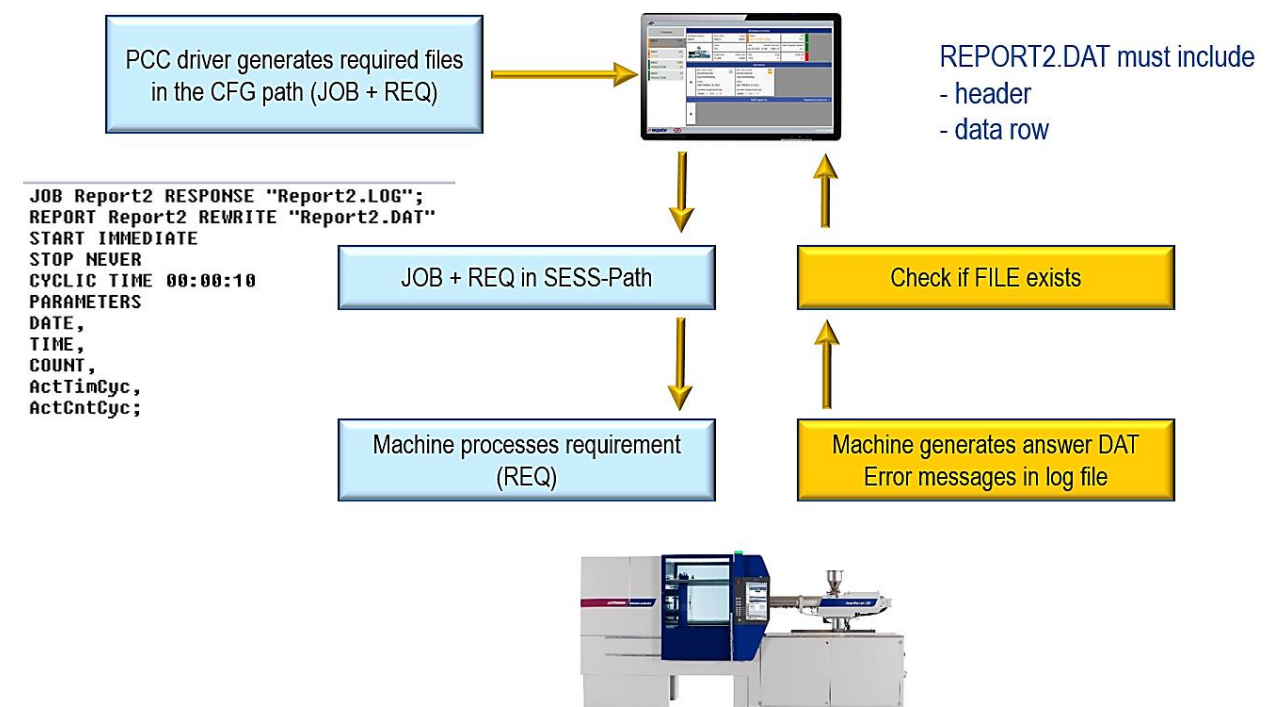


Figure 1: Process description EUROMAP 63 MDE Report2.DAT

In general, the EUROMAP 63 file interfaces know different data types:

- JOB file: defines a job. Either a data request job or a cancel job.

Example "qpath" folder:

```

abortall.job
download.job
getid.job
getinfo.job
report1.org
sess0000.req
sess0001.req
sess0002.req
sess0003.req
sess0004.req
sess0010.req
sess0011.req
upload.job

```

- REQ file: starts a job.

Example "sesspath" folder:

```

ABORTPDV.JOB
ABORTPDV.LOG
REPORT2.JOB
report2.tmp
SESS0000.bak
SESS0004.bak
SESS0007.bak
SESS0007.RSP

```

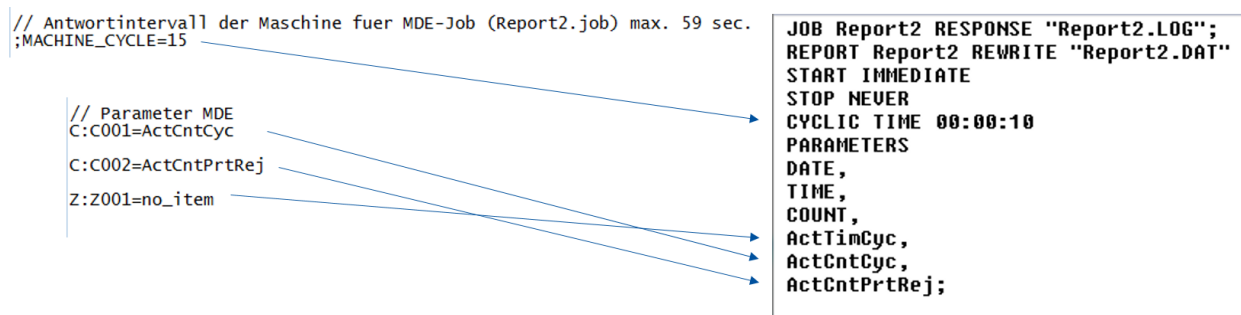


Figure 2: Sample configuration: e63.ini according to the Report2.dat

The MDE and PDV connection via the driver e63.dll. For the driver, the sample configuration file e63.bsp is available. In the PCC directory, rename the e63.bsp to e63.ini. Then change the parameters as required for the machine.



Sample configuration file: „e63.bsp“

```
[SERVICE]
info=E63.dll
intervall=0
testmode=0
tracing=1
TraceLevel=5
ExecuteQueue=1
ThreadBaseId=100

[Maschine1]
loglevel=16
; Source path for JOB files
qpath=C:\MPDV\AIP2\cfg\Maschine1\
; Session-Path for FTP exchange
sesspath=Z:\Maschine1\

; Save the counter, INTERVAL in sec.
SAVE_COUNTER=ON

; Receiving alarm messages via job etCurrentAlarms
;GET_CURRENT_ALARMS=ON

; Response Interval of the machine max. 59 sec.
MACHINE_CYCLE=15
OVERFLOW=999999
OVERFLOWMODE=0

; Selection of the machine type:
; DEMAG : Demag machine
; KM : Krauss Maffei machine
; ENGEL: Engel machine
; BATTENFELD : Battenfeld machine
Maschinentyp=DEMAG
; Selection of the E63 controller
; CCL912 (DEMAG)
; CIPC (DEMAG)
; DIP12 (DEMAG)
E63-Controller=DIP12

; Timeout to the machine in ms
Timeout=60000
; Timer for processing cyclic of E63 data in ms
Polltime=5000

; Parameter MDE channel configuration: on the left side of the equals sign is the
HYDRA channel config, on the right the e63 interface variables
;C:C001=ActCntCyc
; E63 variable is automatically requested and transferred from the machine control
;Z:Z001=ActTimCyc

; PDV channel
;T:T001=ActCntCyc
;P:P0001=ActPrsXfrHyd[1]
;P:P0002=ActTmpOil
;P:P0003=@CycleTime.sv_dMoldCloseClmpBldPDP
```

3.3 E63 DNC configuration

You use the driver e63_dnc.dll to connect DNC. For the driver, the sample configuration file e63_dnc.bsp is available. In the PCC directory, rename the e63_dnc.bsp to e63_dnc.ini. Then change the parameters as required for the machine.



Sample configuration file: „e63_dnc.bsp“

```
[SERVICE]
info=e63_dnc.dll
intervall=0
testmode=0
tracing=0
TraceLevel=0
ExecuteQueue=1
ThreadBaseId=100

[Maschine1]
; Source path for JOB files
qpath= C:\MPDV\AIP2\cfg\Maschine1\
; Session-Path for FTP exchange
sesspath= Z:\Maschine1\

loglevel=1
;Maschinentyp=DIP12

; Settings for DEMAG
; E63-Controller=DIP12
; Zip=ON
; ZipDir=ON

Timeout to the machine in ms
Timeout=120000
Polltime=10000

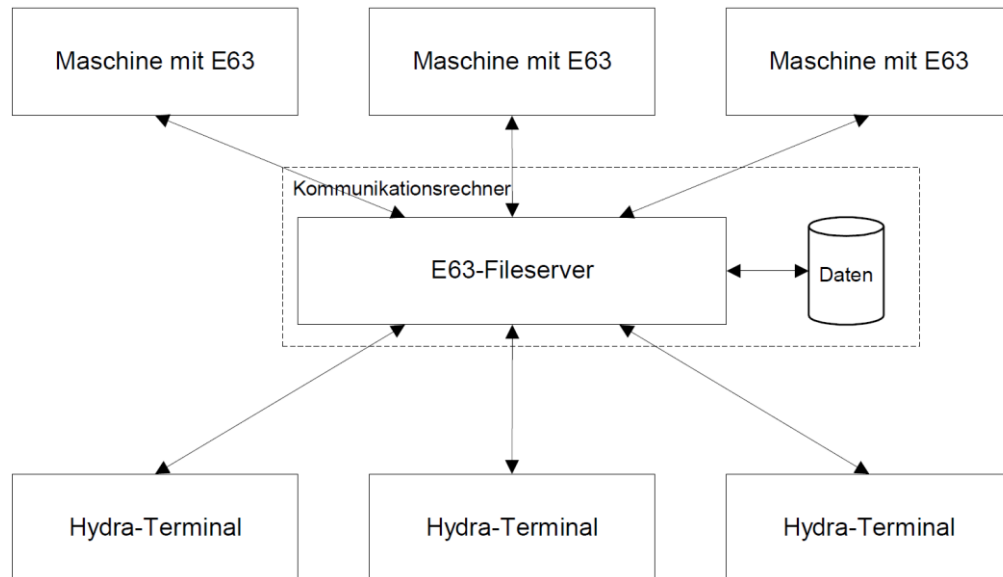
; DNC-UPLOAD-REPORT
sessup=SESS0010

; DNC-DOWNLOAD-REPORT
sessdown=SESS0011

; Name of the DNC JOB files
jobup=UPLOAD
jobdown=DOWNLOAD
D:DNC_500001=D:DNC_ + Maschine1
```

4 Annex

4.1 Architecture



You use the Ethernet interface for the file transfer between machine control and the higher-level computer (i.e. communication device). There is a screen in the control to set the communication parameters (like the TCP/IP address of the host computer and the name of the "session" directory). HYDRA processes the files using Windows file access via the operating system locally or via the network.

4.2 Installation FTP server

4.2.1 General Notes



The FTP server "Internet Information Services" (IIS) of Microsoft cannot process specific FTP commands (e.g. LISTDIR *.*) with some machine manufacturers (e.g. DEMAG). Many machine manufacturers do not support the IIS.

Use alternative products like the FileZilla, for example, or follow the recommendations of the machine manufacturer.

4.2.2 Sample installation FileZilla-FTP-Server



Important: This is only a configuration example. The actual installation must be based on the specifications of the machine manufacturer.

The FileZilla server is a graphic (Windows and Linux) open source FTP server for different platforms. Using the FileZilla server, files can be exchanged quickly and easily between the file directory and the clients via ftp.



You can download the free FileZilla server at: <https://filezilla-project.org>

Follow the installation assistant of the "FileZilla Server". Be careful to "Install as service". If required, unblock the FileZilla server in the firewall.

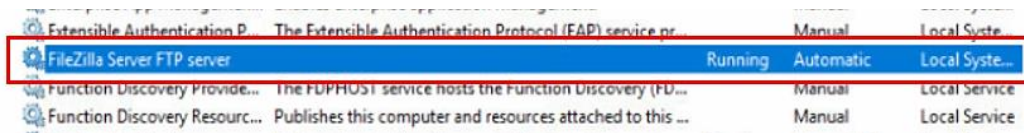


Figure 3: FileZilla FTP server runs as service in the background.

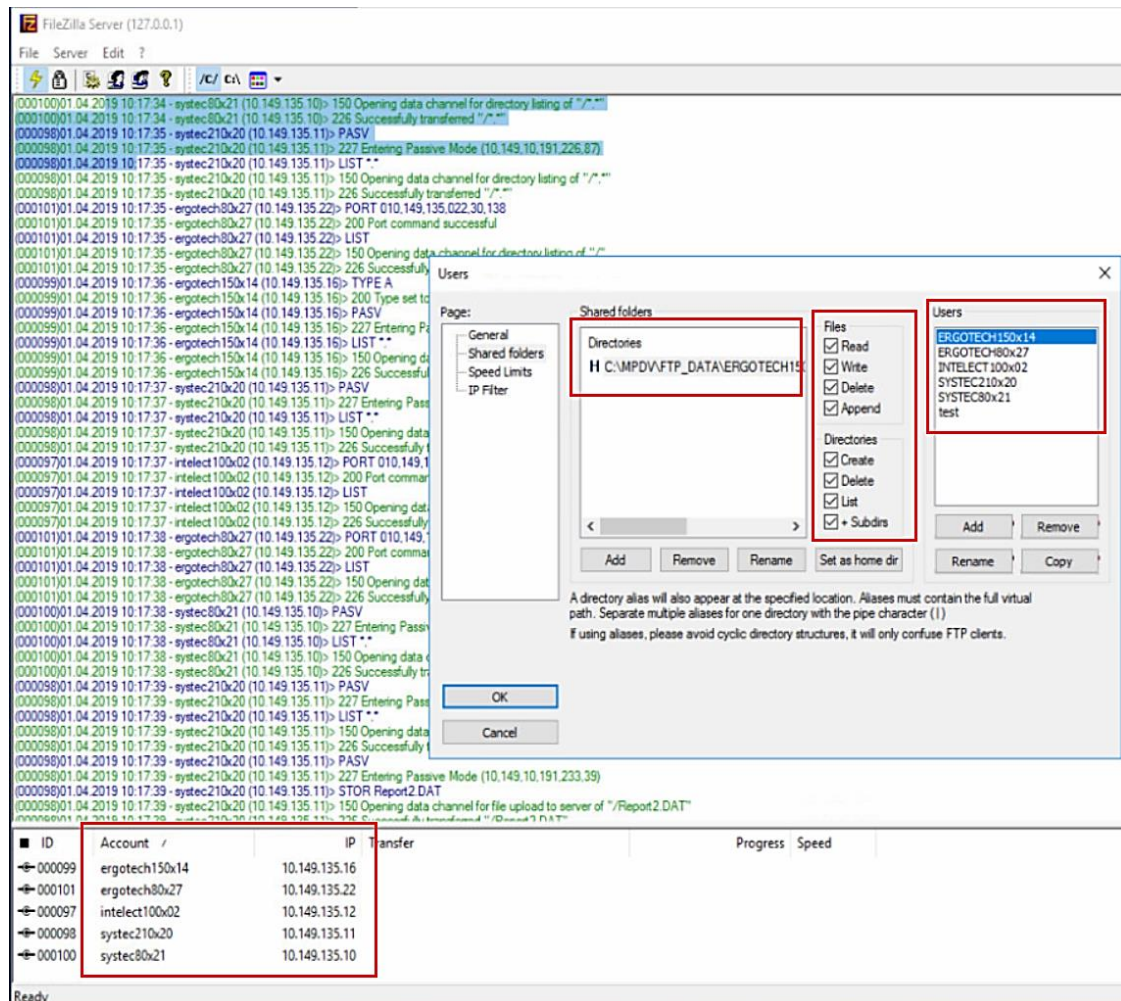


Figure 4: Directory and user configuration FileZilla-FTP-Server

For each machine, an FTP user with an own directory "shared folder" must be created. The user must have the following rights for the server: Files: Read, Write, Append, Delete and Directories: Create, Delete, List, + Subdirs. The search pattern for session files is *.*.

You can monitor the communication using the interface of the FileZilla server. The upper part of the screen of the FileZilla server interface shows the communication between server and client. The lower part shows all active FTP sessions of the clients. Here, all E63 machines must be listed that have an active connection to the FTP server.

4.3 Examples of specific manufacturers



This section describes configurations that are specific to a manufacturer. MPDV cannot control how the EUROMAP63 standard is implemented by the machine manufacturers.

4.3.1 Example DEMAG

The required settings are displayed below. See the screenshot EUROMAP63 (No. 199).

5	199 EUROMAP 63	0.00	!SIMULATION!
	[6] Drahtbruch Werkzeugheizung		13:42:00

Einstellungen Server	
Server Name	pluto
Server Pfad	\
IP-Adresse (Server)	192.168.001.003
Username	sigma
Password	*****
Subnetmask	255.255.255.000
Gateway Adresse	000.000.000.000

Einstellungen Ergocontrol	
Set IP-Adresses	☐
Scantime Server	5
IP-Adresse (eigene)	192.168.001.002
Status:	DIP011: Euromap-Config ready
Fehler:	data channel ok

Setup							
-------	--	--	--	--	--	--	--

Figure 5: EUROMAP63 (No. 199) configuration screen of DEMAG

The settings are only active when the program switch "Set IP address" has been enabled or when the ergocontrol has been booted.

The settings are as follows:

Server	
"Server Name"	Name of the server in the net. Must be an FTP server.
"Server Pfad"	Path name; the ergocontrol regularly uses this path to search for session files. A so-called "Home Directory" of the FTP server can be reached by entering \ into the field.
"IP-Adresse (Server)"	IP ¹ address in format zzz.zzz.zzz.zzz where 'z' is a number. Not permitted: leading or trailing zeros at the beginning and end of the IP address, e.g. 112.231.001.000. You can get your IP address from your network administrator.
"Username"	Name that the ergocontrol uses to log in to the FTP server.
"Passwort"	Password for the FTP server. Enter the complete password, if required fill with spaces.
"Subnetmask"	Assign according to IP protocol instructions. Refer to your network administrator.
"Gateway-Adresse"	Assign according to IP protocol instructions. You can get your IP address from your network administrator.
Ergocontrol	
	Only use when all other settings are made.
"Scantime Server"	Time interval (in seconds) used by the ergocontrol to request session files from the server.
"IP-Adresse (eigene)"	IP address of the ergocontrol. You can get the IP address for the ergocontrol from your network administrator.
"Status"	Status of the Euromap63 software in the the ergo control; only display, no input option.
"Fehler"	Errors reported by the ergocontrol; only display, no input option.

On the FTP server, set up a user with user name and password as shown above in the ergocontrol screenshot "EUROMAP 63".

Depending on the DEMAG E63 controller, a blank line is inserted between the header row and the value rows for the E63 response (job file).

Example.: REPORT1.DAT

```
DATE,TIME,COUNT,ActTimFill[1],ActTimPlst[1],ActVolCsh[1],ActVolPlst[1],ActVolXfr
[1]
```

BLANK LINE

```
20061107.14:17:11,39,0,5,8,52.65,10.60
```

BLANK LINE

E63 controller **with** blank line:

- CCL912
- CIPC

E63 controller **without** blank:

- DIP 11
- DIP 12

4.3.2 Example Krauss-Maffei MC4/MC5

Before communication can be performed, you must configure the interface in the machine control.

ACCESS RELEASE:

To access the Euromap63/SPI image pages, proceed as follows:

Password level 1 or 4 must be set with permission for data input.

Press the following keys to open the page for the settings of the Euromap63-/SPI interface.

Key  then soft key 

Enter the IP address of the machine here.

Then use the soft key

 to configure the Euromap interface.

Figure 6: EUROMAP63 configuration screen of Krauss-Maffei

This page displays the data for communication with the server host (FTP server), the settings for the data transferred and diagnosis results for the interface operation.

IMPORTANT: If you must change data entered, you must restart the control to enable the changed data.

The following specifications are required for the communication with the server host and its FTP server program:

TCP/IP address of the server host ("TCP/IP-Adresse")

This is the address of the central host where the FTP server program is running.

Name of the server host ("Hostrechner-Name")

Each server host must get a name. This version does not support a name server. Any name is therefore possible.

User name for the FTP program ("Benutzername")

This is the login name used for the machine in the FTP server.

Password for the FTP program ("Passwort")

This is the password used in the FTP server.

The password is not hidden, because the access area is restricted as specified by the FTP server.

Directory in the server host where the Euromap63-SPI session files are stored ("Verzeichnis der SESSION-Dateien").

This is the directory where the machine searches for the session file (command file to the machine). The following directory names are possible:

<letter of the drive>:\<directory name> for Windows NT Server

<directory name> relative to the login directory

\ Login directory = server directory

Euromap63 (on/off/with monitoring - "ein/aus/mit Überwachung")

Use this to log on and off the activated Euromap63 interface or to operate it with an error display. If communication to the FTP server is interrupted, a window opens showing the operator a communication breakdown.

Set parameter dimensions ("Parameterdimensionen setzen"):

There are two options:

SI units SI units are the basis for the parameter values in the MC4 control. If the SI units are selected to provide data for the central server, you can choose any other format for the display on the machine.

Times	in sec
Pressures	in bar, hydraulic pressure
Distances	in mm
Speeds	in mm/s
Revs	in 1/min
Forces	in N

Display format If display format ("Anzeigeformat") is selected, then the current display setting (e.g. US dimensions) is used for data output. You must not change the display setting.

IMPORTANT If the display setting is changed, all parameters for dimensions cannot be read or changed via the interface. The central server receives the error message "Parameter not found".

Set the current text language for Euromap63/SPI ("Textsprache")

With the selection of the language, the current language is used as language of the parameter names for the interface.

Proceed as follows to change the language:

1. Select the required language for the display.
2. Return to this page.
3. Click the option "Set ... as current language" ("Aktuelle Textsprache ... setzen").

The parameter names are then transferred in the selected language for the Euromap interface, independent of the language currently displayed in the machine control.

The button then shows the letter symbol for the set language (e.g GB for English).

IMPORTANT The communication is not started if the parameter "language" is not selected. On a further page, you can make additional specifications for the Euromap63 interface.



Figure 7: Configuration screen of Krauss-Maffei: setting for the time delay

Time delay for query ("Zeitverzögerung zur Abfrage"): The FTP server is permanently prompted for new commands. The time delay specifies the time in seconds between the prompts.

FTP server with minimum functions ("FTP-Server mit minimalen Funktionen"): The FTP server does not use directory functions.

FTP with command CD ("FTP mit Kommando CD"): Some FTP servers require the command CD with directory name before they can call a directory.

Using stored measured values ("Verwenden von gespeicherten Messwerten")

Usually, the current measured values are requested during the change of the cycle counter. The stored measured values are issued with a delay of 5 cycles when the function is activated. The reason for the delay is that the values can be assigned to the correct cycle, for example if a scale is connected.

Saving and loading the configuration ("Konfiguration sichern/Konfiguration laden")

You can save the settings for the Euromap63 interface on a disc and re-import them. When you save the settings on the disc, the logbook of the Euromap63 interface is also transferred and can be viewed on the PC.

4.4 Typical E63 variables

Acronym	Description
ActCntCyc	Shot counter Total
ActTimCyc	Cycle time
ActVolCsh[1]	Melt cushion
ActTimFill[1]	Injection time
ActTimPlst[1]	Dosing time / plastication time
ActPrsMachSpecMax	Change cavity pressure
ActPrsXfrSpec[1]	Changeover pressure spec.
ActVolXfr[1]	NDR - Position (PDE), per injection unit
ActVolPlst[1]	Dosing stop (PDE), per injection unit
ActTmpBrlZn[1,1]	Cylinder heating zone 0 (EF)
ActTmpBrlZn[1,12]	Cylinder heating zone 11 (DH1)
ActTmpBrlZn[1,2]	Cylinder heating zone 1 (MH1)
ActTmpBrlZn[1,3]	Cylinder heating zone 2 (MH2)
ActTmpBrlZn[1,4]	Cylinder heating zone 3 (MH3)



The available E63 variables are specific for each machine. You can use the machine query "getid" to identify the available machine-specific variables.